



Python Project 4 - Polynomial Regression

Are we getting taller? There is no noticeable height change in humans from when record keeping started to the early 1800's but since that time the average height has been increasing. Today, on average, we are taller than our predecessors but these gains vary considerably from country to country. Researchers found that today Dutch men have the highest average heights at 182.5 centimeters (about 6 feet) and Latvian women at 170 cm (about 5 feet 6.5 inches) with Dutch women in second place. A century ago the United States ranked third in adult male height but it currently ranks 37th so the gain here is not as large as in other countries.

We want to use linear regression to model the growth in the average height of men or women (your choice) in the United States and the Netherlands.

The data for this lab was taken from the website ourworldindata.org which got the data from NCD RisC (a network of health scientists around the world).

The data is contained in the file `human_heights.txt` which consists of 5 columns of numerical data in the following order: year, Dutch men average height, Dutch women average height, US men average height, US women average height. The heights are given in centimeters.

1. Choose to compare either Dutch men to US men or Dutch women to US women
2. Import libraries
3. Enter the recorded values for comparison that I have provided
4. Read in data from file `human_heights.txt`; you can either put your data in 1D arrays or in a matrix. Plot your data for Dutch and US on same plot; add labels to axis, plot title and legend. To add a legend use the label option on `plt.plot` and the command `plt.legend()`.
5. Use linear regression to fit the data for either men or women; plot both Dutch and US results on the same plot; add labels to axis, plot title and legend.
6. Calculate the variance
7. Use regression to fit the data for either men or women using a parabola; plot both Dutch and US results on the same plot; add labels to axis, plot title and legend.
8. Calculate the variance
9. Decide which fit (a line or a parabola) is better; justify your answer quantitatively. Then use whichever is better to predict the heights in 1955 and 1995; calculate the percent error in these predictions and the exact values I provided. The percent error should be positive and found by taking the absolute value of the difference in the predicted and actual heights divided by the actual height times 100 (to get percent). Are the predictions at different years comparable?

Indicate here whether you are comparing Dutch men to US men or Dutch women to US women

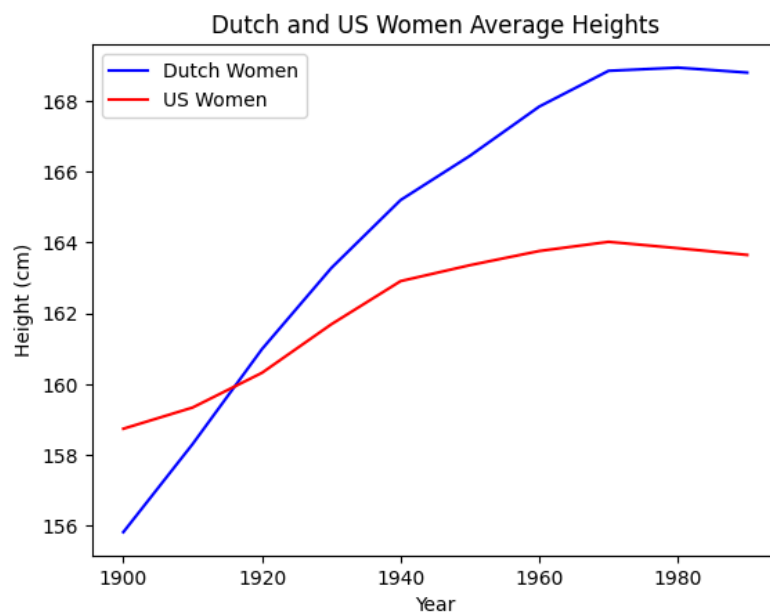
```
# Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Input exact values for comparison
#
dutch_man_1925 = 174.83; us_man_1925 = 174.53
dutch_man_1955 = 180.23; us_man_1955 = 177.22
dutch_man_1995 = 182.54; us_man_1995 = 177.16
#
dutch_woman_1925 = 162.2; us_woman_1925 = 160.97
dutch_woman_1955 = 167.11; us_woman_1955 = 163.54
dutch_woman_1995 = 168.73; us_woman_1995 = 163.56
```

```
# Read in the data and plot; add axes labels, plot title and legend
year = np.loadtxt('human_heights.txt', skiprows=2, usecols=0, unpack=True)
dutch_women = np.loadtxt('human_heights.txt', skiprows= 2, usecols= 2, unpack=True)
us_women = np.loadtxt('human_heights.txt', skiprows=2, usecols=4, unpack=True)

plt.plot(year, dutch_women, 'b-', label='Dutch Women')
plt.plot(year, us_women, 'r-', label='US Women')
plt.xlabel('Year')
plt.ylabel('Height (cm)')
plt.title('Dutch and US Women Average Heights')
plt.legend()
plt.show()

# Note: to add legend use label option in plt.plot (e.g., label='Dutch Men') and then use
# the command plt.legend ()
```



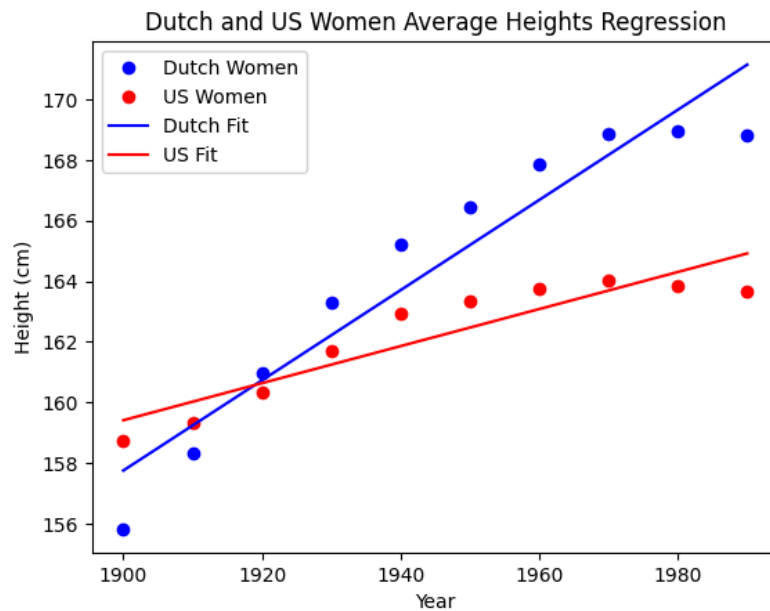
```
# Linear regression fit for both Dutch and U.S.; plot and print out the line
coef_dutch = np.polyfit(year, dutch_women, 1)
coef_us = np.polyfit(year, us_women, 1)

print(coef_us)
print(coef_dutch)

dutch_fit = np.poly1d(coef_dutch)
us_fit = np.poly1d(coef_us)

plt.plot(year, dutch_women, 'bo', label='Dutch Women')
plt.plot(year, us_women, 'ro', label='US Women')
plt.plot(year, dutch_fit(year), 'b-', label='Dutch Fit')
plt.plot(year, us_fit(year), 'r-', label='US Fit')
plt.xlabel('Year')
plt.ylabel('Height (cm)')
plt.title('Dutch and US Women Average Heights Regression')
plt.legend()
plt.show()
```

```
[ 0.06112121 43.28224242]
[ 0.14876364 -124.89727273]
```



```
# Calculate the variance for each fit; use the function that we wrote in a previous notebook
# Input: the x and y arrays for the data points, coefficients of line found using LR
def variance(x, y, coef):
    y_pred = np.polyval(coef, x)
    return np.sum((y-y_pred)**2)/len(y)

# Output: variance

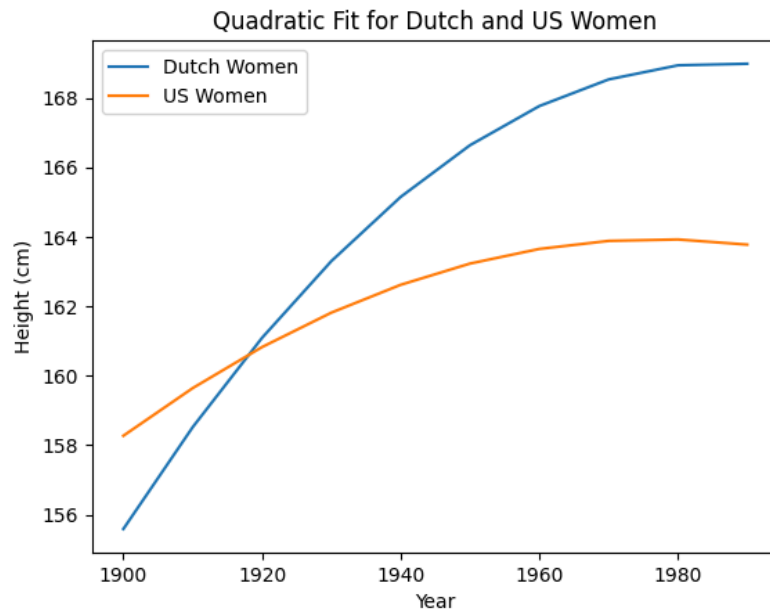
print(variance(year, dutch_women, coef_dutch))
print(variance(year, us_women, coef_us))
```

```
1.743654909090882
0.5499438787878755
```

```
# Quadratic regression fit for Dutch and U.S.; plot and print out the parabolas
coef_quad_dutch = np.polyfit(year, dutch_women, 2)
coef_quad_us = np.polyfit(year, us_women, 2)

dutch_quad_fit = np.poly1d(coef_quad_dutch)
us_quad_fit = np.poly1d(coef_quad_us)

plt.plot(year, dutch_quad_fit(year), label='Dutch Women')
plt.plot(year, us_quad_fit(year), label='US Women')
plt.xlabel('Year')
plt.ylabel('Height (cm)')
plt.title('Quadratic Fit for Dutch and US Women')
plt.legend()
plt.show()
```



```
# Calculate variance for the quadratic fits
def variance_quadratic(x, y, coef):
    return np.mean((y-np.polyval(coef, x))**2)
variance_quad_dutch=variance_quadratic(year, dutch_women, coef_quad_dutch)
variance_quad_us=variance_quadratic(year, us_women, coef_quad_us)

print('Variance for Dutch Women:', variance_quad_dutch)
print('Variance for US Women:', variance_quad_us)
```

```
Variance for Dutch Women: 0.028612181818184083
Variance for US Women: 0.07380380303033693
```

```
# Use best fit to predict average heights in 1955 and 1995 for both Dutch and U.S.; compute percent error;  
# round values to 2 decimal places  
  
pred_dutch_1955 = round(dutch_quad_fit(1955), 2)  
pred_dutch_1995 = round(dutch_quad_fit(1995), 2)  
pred_us_1955 = round(us_quad_fit(1955), 2)  
pred_us_1995 = round(us_quad_fit(1995), 2)  
  
# Percent error calculations  
pct_err_dutch_1955 = round(abs((pred_dutch_1955 - dutch_woman_1955) / dutch_woman_1955 * 100), 2)  
pct_err_dutch_1995 = round(abs((pred_dutch_1995 - dutch_woman_1995) / dutch_woman_1995 * 100), 2)  
pct_err_us_1955 = round(abs((pred_us_1955 - us_woman_1955) / us_woman_1955 * 100), 2)  
pct_err_us_1995 = round(abs((pred_us_1995 - us_woman_1995) / us_woman_1995 * 100), 2)  
  
# Print results  
print("Predicted Dutch height in 1955:", pred_dutch_1955, "cm", "Percent Error:", pct_err_dutch_1955, "%")  
print("Predicted Dutch height in 1995:", pred_dutch_1995, "cm", "Percent Error:", pct_err_dutch_1995, "%")  
print("Predicted US height in 1955:", pred_us_1955, "cm", "Percent Error:", pct_err_us_1955, "%")  
print("Predicted US height in 1995:", pred_us_1995, "cm", "Percent Error:", pct_err_us_1995, "%")  
  
# The quadratic model provided the best fit as seen above in the variance  
# calculations. The percent error was also lower for us height in 1995 by  
# around a percentage point. The quadratic model is able to better fit the data.
```

```
Predicted Dutch height in 1955: 167.24 cm Percent Error: 0.08 %  
Predicted Dutch height in 1995: 168.87 cm Percent Error: 0.08 %  
Predicted US height in 1955: 163.46 cm Percent Error: 0.05 %  
Predicted US height in 1995: 163.63 cm Percent Error: 0.04 %
```